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Institute of Management & Career Courses (IMCC)

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DEPARTMENT OF COMPUTER APPLICATIONS (MCA)

CAPSTONE PROJECT SCOPE PROPOSAL & TECHNICAL SYNOPSIS

Academic Cohort: **2026-2028** • Class: **FY MCA** • Section: **Division B** • **Group 65**

OFFICIALLY APPROVED

Project Title	**Code_Morph**		
Team Name	**Team Marvel Arc** Group 65	**Project Track / Domain**	Artificial Intelligence & Web Development
Project Category	Capstone Team Project	**Submission Date**	6 October 2026
Internal Faculty Guide	**Dr Shilpa Parlikar** ssp.imcc@mespune.in	**External Industry Guide**	**AUTO** Industry Organization

1. STUDENT PROJECT TEAM ROSTER

TOTAL MEMBERS: 2

#	STUDENT FULL NAME	ROLL NO. / PRN	INSTITUTIONAL EMAIL	PROJECT RESPONSIBILITY
1	**Ashutosh M. Uttekar**	2601196	ashutosh110405@gmail.com	Lead **Team Leader (Lead)**
2	**Neel J. Vidar**	2601198	vidarneel15@gmail.com	Core Team Member

2. PROBLEM STATEMENT & ACADEMIC SYNOPSIS

Developers often face difficulties when migrating code between different programming languages or technologies, as manual conversion is time-consuming and can introduce errors. Identifying, understanding, and fixing these errors further increases development effort. CodeMorph aims to simplify this process by converting code, detecting errors, explaining their causes, and providing suitable solutions for efficient software migration and development.

3. PROJECT SCOPE & FUNCTIONAL OBJECTIVES

The main objective of CodeMorph is to simplify code conversion between programming languages by reducing manual effort, detecting and explaining errors, providing suitable solutions, and validating the converted code for accurate and efficient software development.

4. TECHNOLOGY STACK & INFRASTRUCTURE ENGINES

PRIMARY DEVELOPMENT & ARCHITECTURAL STACK

Python HTML5 CSS3 JavaScript Flask

DATABASES, CACHES & STORAGE ENGINES

MySQL

5. ARCHITECTURAL MODULE BREAKDOWN & DELIVERABLES

4 DEFINED MODULES

#	FUNCTIONAL MODULE	DETAILED SYSTEM SPECIFICATION & DELIVERABLE SCOPE
M1	**Module 1: Code Conversion Module**	Converts source code from one programming language to another.
M2	**Module 2: Error Detection & Debugging Module**	Detects errors, identifies their causes, and provides suitable solutions.
M3	**Module 3: Virtual IDE Module**	Provides an online coding environment where users can enter prompts or requirements and generate code in their desired programming language through an API.
M4	**Module 4: User & History Management Module**	Manages user accounts, saved code, conversion history, and feedback.

6. TANGIBLE DELIVERABLES & EXPECTED OUTPUT

The expected outcome of Code_Morph is to provide a unified development environment that makes programming tasks faster, easier, and more accessible. The system will help users reduce development time, improve code quality, understand programming logic, and minimize errors during development.

FACULTY SUPERVISOR REVIEW & AUDIT ASSESSMENT:

*Scope proposal submitted by student team. Initial technical evaluation completed by internal review committee. Status: **OFFICIALLY APPROVED**.*

Ashutosh M. Uttekar	**Dr Shilpa Parlikar**	**AUTO**	**Head of Department**
Lead Sign-off / Date	Internal Faculty Supervisor	External Industry Mentor	Dept. of Computer Applications (MCA)

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